



1. Update logs

Update instructions	Update date	
First released version	2021-03-05	
Merged interface	2021-03-29	
Removes list workNo field, adds Remote upgrade instruction	2021-04-21	
Added QR code recognition Settings and added visitor list access time Settings	2021-06-10	
Cloud notes added	2021-11-17	
Added time period control interface	2021-12-07	
Record Upload Adds health code field	2021-12-28	
Pre-setup instructions added to the attachment	2022-04-13	
Added remote restart interface	2022-06-11	
Added interface for re-pushing device records	2022-07-02	
Note: The device has repaired the personnel validity period. If the platform issues the personnel validity period, it must be updated (check the [cloud access equipment personnel information] interface).	2022-07-21	
Set Temperature parameter interface, Get/Set Basic Device Parameter interface Added configuration parameters	2022-11-14	
Added online comparison interface (please through http://47.104.64.115:9001/ for new updates)	2023-02-17	
Added custom voice return to online alignment interface	2023-05-12	
Added the interface for obtaining device personnel information and adjusted the document directory	2023-08-18	

2 Document object-oriented

The object orientation of this document is as follows:

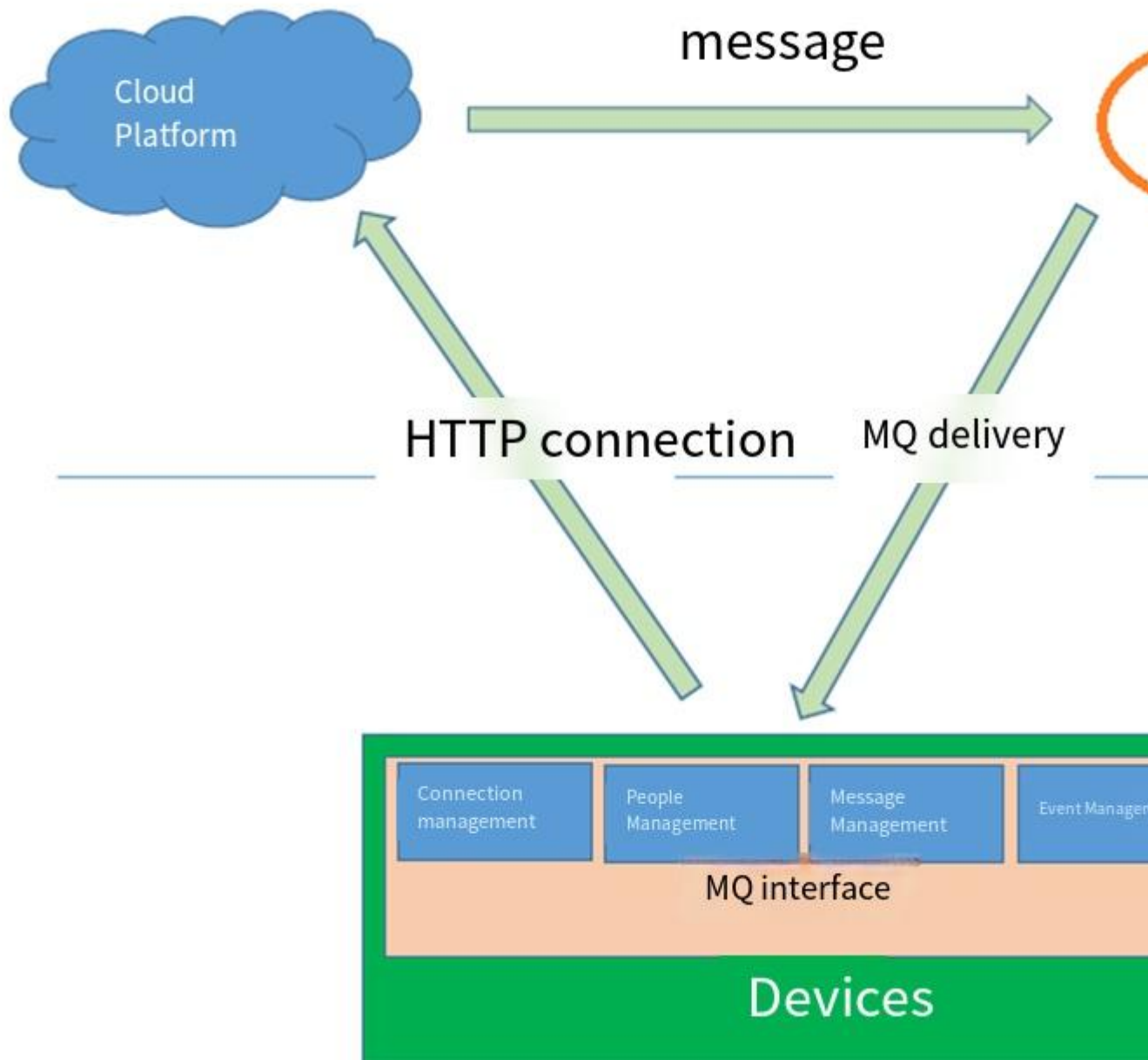
Categories for readers	Purpose
Technician	Know the general content of the interface to which the device is interfacing
System Developer	Understand the device docking interface and conduct development docking

3. Architecture description

3.1 Architecture design of end cloud system based on MQ

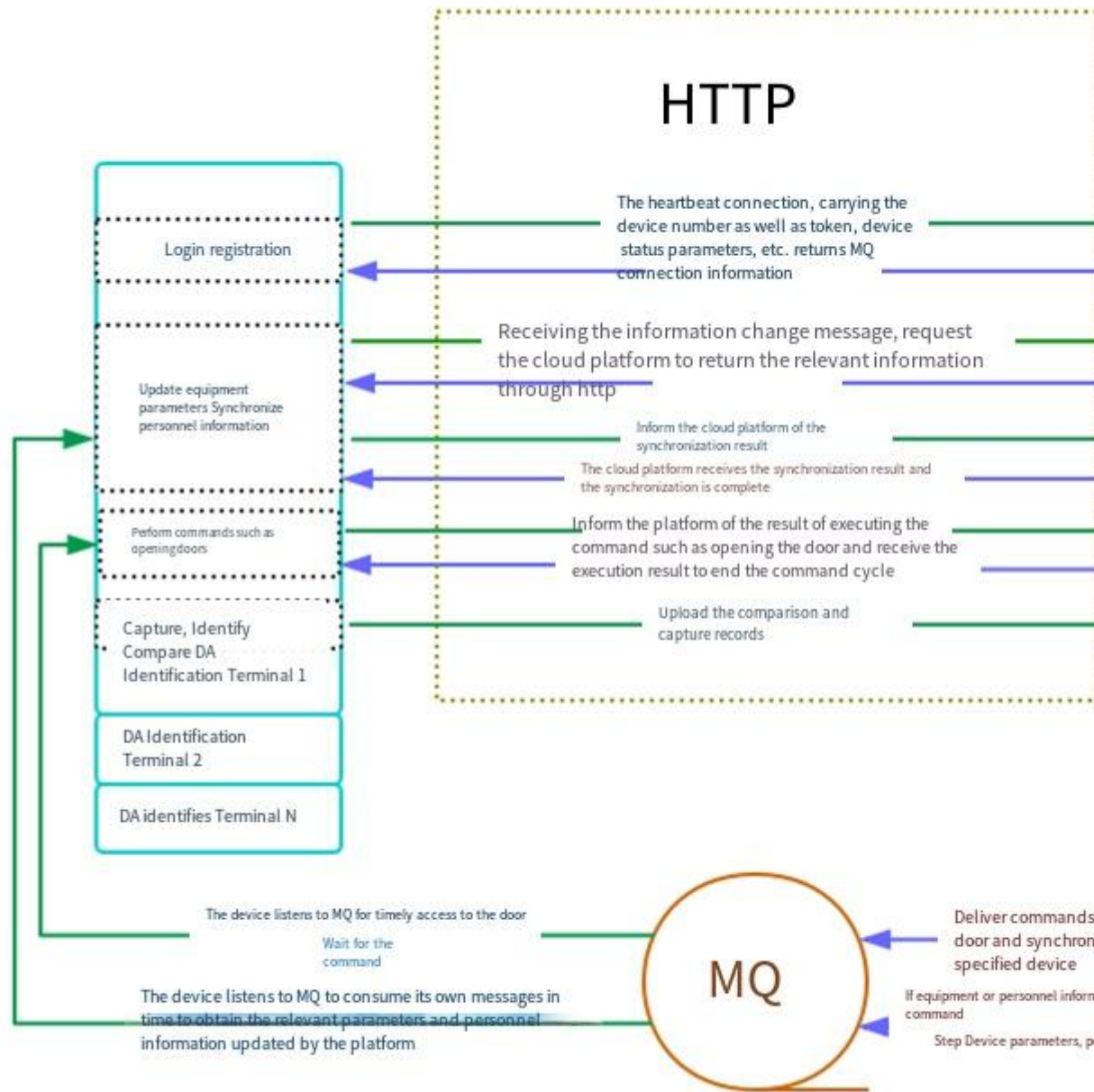
- The terminal device provides a set of interface protocols, which are used to regulate the messages, comparison events, personnel information and other behaviors between the cloud platform and the terminal device. The detailed system framework is as follows:
- **Note:** The cloud platform involved in this document refers to the cloud platform of the reading party (the peer party), and all interfaces involved in this document are the platforms of the device directly connected to the reading party (the peer party).

The MQ server in the architecture needs to be built by the reading party (the receiving party).



Support MQTT(EMQX)

3.2 Communication between Devices and Platforms



MQTT(EMQX)

4. The cloud sends instructions

4.1 Agreement



- The platform delivers information to the device via MQTT, for which the device has dev_sno as topic.
- The message body is a json string

The device recognizes the command according to method

5 Log in to the device

Description

- Interface address: Perform login, personnel information acquisition, identification upload and other information communication server interface address
- Device registration number: can be configured on demand, used for multi-company platform, distinguish the company, region and other information of the device, if there is no platform that requires this can be filled in arbitrarily.
- The interface address and device registration number can be configured in the device configuration
- After the device is started, the login request is executed through the configured: interface address + configured registration number and device serial number. The subsequent operations can be performed only after the token is successfully obtained.
- The device is automatically registered upon the first login.
- The platform verifies that the device login is legitimate and the legitimate device returns mqtt connection information and token
- Only legitimate devices will get the token as a timeout check and permission verification
- The token is used by the platform to determine the validity of the device. The platform can update the token periodically. For platforms without this requirement, the same token can always be used.
- Heartbeat connection: Devices log in to the platform every 30 seconds, maintain communication, and report relevant status parameters. Devices that maintain heartbeat connections.

Port: 5389 port can be customized

Request URI

- `/device/login`

How to request

- POST

Parameter types



- application/json

Parameters

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device number (device MAC address)
sign_type	is	string	Signature type (fixed dev_sign)
register_no	is	string	Device registration number (not yet meaningful)
type	is	int	Type of equipment: 1 Access control machine
name	is	string	Device name
throughType	is	string	Entry and exit, 1: out 2: in
threshold	is	float	Recognition threshold [0.0-100.0]
detectAlive	is	string	close: Live detection is not enabled. IRAlive: IRAlive mode
openIRWindow	is	bool	Whether to open IR small window
retryInterval	is	float	Face recognition deduplication time, in seconds
isFirst	no	bool	Whether to log in for the first time. This parameter is true if you switch to another platform account midway through the login

Sample Request

```
{
  "detectAlive": "IRAlive",
  "dev_sno": "FF:0C:F4:12:67:05",
  "Device_ip" : "192.168.1.135",
  "distance": 4,
  "isFirst": false,
  "name": "Terminal",
  "openIRWindow": false,
  "register_no": "12",
  "retryInterval": 2,
  "sign_type": "dev_sign",
  "threshold": 70,
  "throughType": "2",
  "type": 1
}
```

```
}

```

The device uses the MQTT protocol and the cloud returns the sample

Parameter name	Required	Type	Instructions
code	is	int	0: Success Non-0: failure
dev_sno	is	string	Device number (device MAC address)
msg	is	string	Log in successfully
success	is	bool	true
token	is	string	
mqinfo	is	object	mqtt connection information
mqinfo.host	is	string	
mqinfo.keepalive	is	int	
mqinfo.login	is	string	
mqinfo.password	is	string	
mqinfo.port	is	int	Port
mqinfo.qos	is	int	
mqinfo.topic	is	string	MAC address of the device

[Back to Example](#)

```
{
  "code":0,
  "msg":" Login successful ",
  "success":true,
  "dev_sno":"0A:0C:F4:12:67:05",
  "token":"eyJkZXZpY2V",
  "mqinfo":{
    "host":"8.xx.1x.xx",
    "keepalive":60,
    "login":"admin",
    "password":"admin",
    "port":1883,
    "qos":1,
    "topic":"FF:AA:F4:12:67:05"
  }
}
```

```
}
```

See Attachment I for a way to configure your login address

After the device requests the platform through http, the platform returns the mq login address

Can't log in MQTT after setting? Please strictly follow the above field json detection, and through the mqtt tool to check whether the mqtt server information is incorrect

6. List management interface

Type of person identity

person_type The corresponding person type

value	Type	Instructions
0	string	TBD
1	string	Visitor (The person who is set as a visitor when the person is delivered is defined as a visitor, and the person who is not recorded on the device is defined as a visitor by direct human witness comparison)
2	string	Stranger
4	string	Whitelist (staff)
5	string	Blacklist

6.1. The cloud sends synchronizer instructions

Purpose: After the cloud personnel is added or modified, send the synchronization personnel command to the device.

Each group sends 10 people, do not send all the platform personnel to the device through one group at a time

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value sync_person
data	is	object	Data



notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- data Parameter Description

Parameter names	Required	Type	Instructions
path	is	string	Request the cloud address of the personnel information interface
path_params	is	object	Request the parameters that the interface needs to carry to get people information

- path_params Parameter description

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
limit	is	int	Number of people to synchronize per instruction (It is recommended to issue 10 people each time, and then issue the next batch of people until the issue is completed; The platform can also directly issue all personnel at one time, but due to the long delivery time, it will block the entire mqtt and http other instructions, which will directly lead to the device login interface will not continue to request until the completion of the login interface request continues, the same as other instructions (that is, a large batch of delivery will lead to the device no longer request the login interface until the completion of the delivery))
offset	is	int	Number of people synchronized
total	is	int	The total number of people to synchronize
person_list	is	string	List of person ids
person_type	is	string	Personnel type

- Examples

```
{
  "method": "sync_person",
  "data": {
    "Path" : "http://192.168.1.16:5389/device/sync_person",
    "path_params": {
      "dev_sno": "04:12:53:FA:21:DA",
      "limit": 10,
      "offset": 0,
      "total": 1,
      "person_list": ["99064", "99067"],
      "person_type": "4"
    }
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device request **data.path** for personnel information:

- Interface description: The device requests all personnel information for the batch person_id
- Request method: **POST**
- Content-Type: **application/json**

```
{
  "dev_sno": "04:12:53:FA:21:DA",
  "path": "http://",
  "path_params": {
    "dev_sno": "04:12:53:FA:21:DA",
    "limit": 1,
    "offset": 0,
    "person_list": [
      "452154515",
      "14541564"
    ], // Device to request all personnel information of the personnel id
    in the array
    "person_type": "4",
    "total": ""
  },
  "token": "b96266221d232f04"
```

}

Platform returns personnel information

Parameter name	type	Instructions
person_id	string	Person id
person_name	string	Name
person_type	string	Personnel type, 1 visitor, 4 employee
sex	int	Gender 1 Male 2 Female 0 Unknown
templateImgUrl	array	Template photo (ask the platform developer to do photo compression function, picture size control within 500K, resolution greater than 400*400, less than 1080*1920)
ic_card	string	ic card
qr	string	Qr Code
id_card	string	Id number
birthday	string	Date of birth
throughDateFrom	string	Visitor Access start date (yyyy-MM-dd) Note: Add a space after dd
throughMomentFrom	string	Visitor Access start Time (HH:mm:ss) 24 hours Notice format
throughDateTo	string	Visitor Access Deadline (yyyy-MM-dd) Note: Put a space after dd
throughMomentTo	string	Visitor Access cutoff time (HH:mm:ss) Note the format
TimeGroupId	string	Normal time period (defaults to 0 when platform does not issue this field) 0: Any time can be successfully identified, 1-4 Monday to Sunday, the specified time period can be successfully identified



SpecialTimeGroupId	string	Special Time Period ID
--------------------	--------	------------------------

Platform Return Example

```
{
  "code": 0,
  "msg": "OK",
  "offset": 30,
  "person_list": [{
    "birthday": "",
    "ic_card": "",
    "id_card": "51032119xxxx043681",
    "person_id": "292068",
    "person_name": "xxx",
    "person_type": "4",
    "sex": 2,
    "templateImgUrl": [
      "http://XXXXX.oss-cn-shenzhen.aliyuncs.com/group2/a2fb8cbe5da13a99be662a4845e74578.jpg"
    ],
    "throughDateFrom": "",
    "throughMomentFrom": "",
    "throughDateTo": "",
    "throughMomentTo": "",
    "TimeGroupId" : ""
  }].
  "success": true
}
```

Device requests cloud feedback on synchronization results:

- Interface description: The device feeds back the batch of 10 people into the warehouse, and then the cloud issues the next batch of personnel
- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC

token	is	string	token value
success	is	bool	Success or failure: Whether the personnel data is successfully received this time. false indicates that the platform needs to deliver the batch of personnel data again
method	is	string	sync_person
msg	is	string	Instructions
person_type	is	string	Personnel type
successful	is	string	List of the id of the person who successfully entered the library, string
failed	is	object	id list of people who failed to enter the database
params	is	object	See example, can be empty

- failed Parameter Description

Parameter Name	Required	Type	Instructions
person_id	is	int	Person id
info	is	object	Instructions for failure
info.img_index	is	int	Stencil photo serial number, counting from 0
info.reason	is	string	Reason for failure

- Example

```

1. {
2.   "dev_sno": "test_1234",
3.   "token": "d0e3be35473b2a331f05cc0c7dc6a9b",
4.   "method": "sync_person",
5.   "person_type": "1",
6.   "success": true,

```



```
7. "successful": "[128, 129, 140, 141, 142, 143, 144, 150]",
8. "failed": [
9. {
10. "person_id": 151,
11. "info": [
12. {"img_index": 0, "reason": "No face detected "},
13. {"img_index": 1, "reason": "No face detected "},
14. {"img_index": 2, "reason": "No face detected "}
15. ]
16. },
17. ],
18. "msg": " sync successful ",
19. "params": {
20. "offset": 0,
21. "limit": 10,
22. "total": 58,
23. }
24. }
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

6.2. The cloud sends a personnel deletion command

Purpose: Delete some personnel of the device



Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value delete_person
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved field, notify required params parameter, provided by the cloud service
data.person_list	is	array	List of person ids
data.person_type	is	string	Personnel type

- Examples

```
{
  "method": "delete_person",
  "data": {
    "person_list": [12, 34, 54],
    "person_type": "4"
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Note: It takes a long time to delete a large number of personnel. In this case, the device may not feedback the execution result
- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
----------------	----------	------	--------------



dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "delete_person",
  "success": true,
  "msg": "Success ",
  "params": {}
}
```

- Cloud returns:

Parameter name	Required	type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
```



```

"code": 0,
"msg": "OK",
"success": true
}

```

6.3. Cloud delete all personnel

Purpose: Delete all personnel from the device

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction with the value delete_all
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```

{
    "method": "delete_all",
    "data": {},
    "Notify" : "http://192.168.1.16:5389/device/notify",
    "params": {}
}

```

Device requests feedback from the cloud on execution results:

- It takes a long time for the device to remove a large number of people, in which case the execution results may not be reported back.
- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
----------------	----------	------	--------------



dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "delete_all",
  "success": true,
  "msg": "Delete successfully ",
  "params": {}
}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
```

```

"code": 0,
"msg": "OK",
"success": true
}

```

6.4. The cloud sends a consistency check command

Purpose: When the platform detects that the number of equipment and platform personnel is inconsistent, this command is issued, and the device will report the id of the personnel to the platform for further processing

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction with the value check_consistency
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

data parameters:

Parameter name	Required	Type	Instructions
data.path	is	string	The device accesses this interface and reports the personnel id

- Example

```

{
  "method": "check_consistency",
  "data": {
    "Path" : "http://192.168.1.16:5389/device/check_consistency"
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {
    "dev_sno": " 04:12:53:FA:21:DA",
    "token": "38f8cc4bc6fb4e82c78175e2d56",
    "params": {

```

```

    "personInfo": [
      {
        "person_type": "4",
        "person_list": [
        ]
      },
      {
        "person_type": "4",
        "person_list": [
        ]
      }
    ]
  }
}

```

Device request data.path Report person id:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device sn
token	is	string	token value
params	is	object	Personnel information

params parameters:

Parameter name	Required	Type	Instructions
params.personInfo	is	array	Personnel information
params.personInfo.person_type	is	string	Personnel type
params.personInfo.person_list	is	array	List of person ids.

- Examples

```
{
  "dev_sno": "04:12:53:FA:21:DA",
  "params": {
    "personInfo": [
      {
        "person_list": [
          "10666",
          "100555",
          "10022",
          "10001"
        ],
        "person_type": "0"
      },
      {
        "person_list": [

        ],
        "person_type": "1"
      },
      {
        "person_list": [

        ],
        "person_type": "2"
      }
    ]
  },
  "token": "eyJkZXZpY2VfaWQiOjE2MTQsImRldmljZV9t"
}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions



success	is	bool	Platform receive success or failure
---------	----	------	-------------------------------------

- Example

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

6.5. The cloud gets the list of equipment personnel

Purpose: The cloud obtains the list of people on the face device

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value get_person_all
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- data Parameter Description

Parameter names	Required	Type	Instructions
path	is	string	Request the cloud address of the personnel information interface

Example of command delivery:

```
{
  "method": "get_person_all",
  "data": {
    "Path" : "http://192.168.1.16:5389/device/get_person_all"
  }
}
```



```
    },  
    "params":{  
    }  
}
```

Device request **data.path** to report a list of people:

- Request method: **POST**
- Content-Type: **application/json**

Parameter name	Required	Type	Instructions
dev_sno	is	string	mac
token	is	string	
person_list	is	object	
person_list.person_id	is	array	Person id
person_list.name	is	string	Person's name
person_list.id_card	is	array	Personnel identification number

Example to report:

```
{  
  "dev_sno": " 04:12:53:FA:21:DA",  
  "token": "f9k4l8ds3e92w02d9c82013na0s9",  
  "person_list": [  
    {  
      "person_id": " person ID",  
      "name": " Name ",  
      "id_card": " ID number"  
    }, {  
      "person_id": " person ID",  
      "name": " Name ",  
      "id_card": " ID number"  
    }  
  ]  
}
```

```
}

```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Back to Example

```
{
    "code": 0,
    "msg": "OK",
    "success": true
}
```

6.6. Get equipment personnel details in the cloud

Purpose: The cloud issues the specified personnel id to obtain the details of the corresponding personnel on the face device

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction, with the value get_person
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service



- data Parameter Description

Parameter names	Required	Type	Instructions
path	is	string	Request the cloud address of the personnel information interface
person_id	is	string	The id of the person to query

Example of command issuance:

```
{
  "method": "get_person",
  "data": {
    "Path" : "http://192.168.1.16:5389/device/get_person_all",
    "person_id": "1231231",
  },
  "params": {
  }
}
```

Device request **data.path** to report personnel details:

- Request method: **POST**
- Content-Type: **application/json**

Parameter name	Required	Type	Instructions
dev_sno	is	string	mac
token	is	string	
person	is	object	
person.person_id	is	array	Person id
person.name	is	string	Person's name
person.id_card	is	array	Personnel identification number
person.birthday	no		Date of birth



person.ic_card	no	IC card number
person.person_type	is	Type of personnel
person.photo	is	Personnel photos
person.throughDateFrom	is	Visitor access start date (yyyy-MM-dd)
person.throughMomentFrom	is	Visitor Access start time (HH:mm:ss)
person.throughDateTo	is	Visitor Access Deadline (yyyy-MM-dd)
person.throughMomentTo	is	Visitor access cutoff time (HH:mm:ss)
person.TimeGroupId	no	Time group id

Example of reporting:

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "f9k4l8ds3e92w02d9c82013na0s9",
  "person": {
    "birthday": "",
    "ic_card": "",
    "id_card": "51032119xxxx43681",
    "person_id": "292068",
    "person_name": "xxx",
    "person_type": "4",
    "sex": 2,
    "photo": "/9j....." ,
    "throughDateFrom": "",
    "throughMomentFrom": "",
    "throughDateTo": "",
    "throughMomentTo": "",
    "TimeGroupId": ""
  }
}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
----------------	----------	------	--------------



code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- [Back to Example](#)

```
{
    "code": 0,
    "msg": "OK",
    "success": true
}
```

7. Device operation information

7.1. Cloud door opening

Purpose: Issue the door opening command to the device, the device opens the gate and other access control

Command parameters:

Parameter name	Required	type	Instructions
method	is	string	Name of the instruction with the value open_door
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- [Example](#)

```
{
    "method": "open_door",
```



```
"data": {},  
"Notify" : "http://192.168.1.16:5389/device/notify",  
"params": {}  
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{  
  "dev_sno": "JWSZ1209323212",  
  "token": "38f8cc4bc6fb4e82c78175e2d56",  
  "method": "open_door",  
  "success": true,  
  "msg": "Door opened successfully ",  
  "params": {}  
}
```

- Cloud returns:



Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- [Back to Example](#)

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

7.2. Set system time in the cloud

Purpose: Set the device time

The device will reboot after setting

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value sync_time
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service
data	is	object	data
data.time	is	string	System time yyyy-MM-dd hh:mm:ss. Example: "2020-01-01 12:00:00"

- [Example](#)

```
{
  "method": "sync_time",
  "data": {
    "time": "2020-01-01 12:00:00"
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request mode: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": "JWSZ1209323212",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "sync_time",
  "success": true,
  "msg": "Setup success ",
  "params": {}
}
```

```
}

```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non-0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Back to Example

```
{
    "code": 0,
    "msg": "OK",
    "success": true
}
```

7.3. Reset device parameters in the cloud

Purpose: Restore device Settings to default values

The device will restart after setting

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction with the value reset_device
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service



- Example

```
{
  "method": "reset_device",
  "data": {},
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": "JWSZ1209323212",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "reset_device",
  "success": true,
  "msg": "Setup success ",
  "params": {}
}
```



```
}

```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
    "code": 0,
    "msg": "OK",
    "success": true
}
```

7.4. Set temperature measurement parameters in the cloud

Purpose: To set the related parameters of device temperature measurement. **This interface parameter has been merged into set_basic_settings, but this interface is reserved for use**

The device will restart after the setting

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value sync_TMPParams
data	is	object	Data
notify	is	string	The device returns the request address of the execution result



params	is	object	Reserved fields, notify required params parameters, provided by the cloud service
data.warningTemperature	is	string	Temperature alarm temperature
data.openHeatMap	is	bool	Whether to turn on heat map
data.outSideDetectMode	no	int	Temperature measurement using Environment 0: Standard mode 1: Outdoor mode
data.temperatureMode	no	int	Temperature measurement function 0 Unexpected temperature 1 Temperature measurement + Face recognition 2 Temperature measurement only

- Example

```
{
  "method": "sync_TMPParams",
  "data": {
    "warningTemperature": "37.3",
    "openHeatMap": false
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value



success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": "JWSZ1209323212",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "sync_TMPParams",
  "success": true,
  "msg": "Setup success ",
  "params": {}
}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

7.5. The cloud obtains basic device Settings

Purpose: To obtain device setting parameters

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value <code>get_basic_settings</code>
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "method": "get_basic_settings",
  "data": {},
  "Notify" : "http://192.168.1.16:5389/device/upload_settings",
  "params": {}
}
```

Device request cloud report parameters:

- Request mode: `POST`
- Content-Type: `application/json`
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions



basicSettings	is	object	Set parameters
---------------	----	--------	----------------

- basicSettings:

Parameter name	Required	Type	Instructions
name	no	string	Device name (default: Terminal)
location	no	string	Installation Location (default: Location)
throughType	no	string	Entry and exit, 1: out 2: in
rebootEnable	no	bool	Timed restart false Disabled, true enabled
rebootTime	no	string	Timed restart time, e.g. 000330 restarts at 3:30 a.m. every day 01xxxx-07xxxx indicates the scheduled restart time from Monday to Sunday. 00xxxx indicates the scheduled restart time every day The device restarts at 2-3 am every Tuesday by default. You can skip this setting unless there are special circumstances
closingDelay	no	int	Opening delay, in seconds (s) [0-255]
systemTime	no	string	Device time, example: 2021-01-01 23:00:00
detectMasks	no	bool	Mask detection
detectAlive	no	string	Detect live, close: Do not turn on live detection, IRAlive: IRAlive mode (Turn on live mode)
openIRWindow	no	bool	Whether to open IR small window
threshold	no	int	Recognition threshold [70,100]
manCardMatchThreshold	no	int	Human witness comparison threshold [30,100]
retryInterval	no	int	Face recognition deduplication time Single identification interval, unit: seconds (s) (default 2S) (same deduplication time)
recogSpaceTime	no	int	Face recognition deduplication time



faceMode	no	int	Face recognition mode, 0 disable 1 Enable 2 Health Code whitelist
certificateMode	no	int	Id recognition mode, 0 Disable 1 person ID recognition (default 1) 2 person ID + whitelist verification 3 Swipe ID only 4 Verify ID
cardMode	no	int	Card recognition mode, 0 disabled 1 Swipe mode 2 Person card mode (default 0)
allowStranger	no	int	Stranger recognition 0 Disable 1 Recognize 2 Recognize door opening
model	no	string	Model number
ip	no	string	ip
wifiMac	no	string	wifi-mac address
ethernetAddress	no	string	Ethernet address
sn	no	string	Device MAC
firmwareVersion	no	string	Firmware version
password	no	string	Device password
platformAddress	no	string	Platform login address (After modifying this address, the device will request this address to log in to the platform)
nucleicTime	no	string	Nucleic acid validity time (Only nucleic acid conditions that meet this requirement can pass) 0 Unlimited, 1 24 hours nucleic acid, 2 48 hours nucleic acid, 3 72 hours nucleic acid, 4 7 days nucleic acid
detectRange	no	int	Recognition distance 0 unlimited,1 0.5 m,2 1 m,2 m]
recogRelay	no	string	Identify open door 0 relay no output 1 relay output (After setting the relay does not output, the remote door opening should be called)
warningTemperature	no	string	Temperature alarm temperature
openHeatMap	no	bool	Whether to turn on heat map



outSideDetectMode	no	int	Temperature measurement using Environment 0: Standard mode 1: Outdoor mode
temperatureMode	no	int	Temperature measurement function 0 Unexpected temperature 1 Temperature measurement + face recognition 2 only temperature measurement
picture	no	string	logo image base64 string, no prefix, see example (this parameter is not available)
bannerEnable	no	int	On ads 0 Off 1 On
bannerString	no	string	Advertising tagline
nameDisplay	no	bool	Identify the result, false does not encrypt, true encrypts
Comparison	no	int	Comparison method 0 Offline comparison (default is offline comparison) 1 Online comparison (After enabling online comparison, the device requests 27. Online comparison identification interface)
passwordOpendoor	no	int	Password door opening function 0 disabled, 1 enabled
doorPassword	no	String	Door opening code (6-digit pure numbers)

- Give an example

```
{
  "basicSettings": {
    "AliveThreshold": 5,
    "IsScaleImage": 1,
    "allowStranger": 1,
    "bannerEnable": 0,
    "bannerString": "",
    "cardMode": 0,
    "certificateMode": 1,
    "closingDelay": 1,
    "detectAlive": "IRAlive",
    "detectMasks": false,
    "detectRange": 0,
```



```
    "doorPassword": "111111",
    "ethernetAddress": "02:02:F4:33:47:05",
    "faceMode": 1,
    "firmwareVersion": "V1.10548.20230108-C907E7-build",
    "healthcodeType": 0,
    "ip": "192.168.1.135",
    "location": "Location",
    "manCardMatchThreshold": 60,
    "model": "C907E7",
    "name": "Terminal",
    "nameDisplay": false,
    "nucleicTime": 0,
    "openHeatMap": false,
    "openIRWindow": false,
    "outSideDetectMode": 1,
    "password": "123456",
    "passwordOpendoor": 0,
    "PlatformAddress" : "http://192.142.101.175:8189/",
    "port": "",
    "rebootEnable": true,
    "rebootTime": "020301",
    "recogRelay": 1,
    "recogSpaceTime": 2,
    "retryInterval": 2,
    "sn": "T99FV18FFFLS",
    "systemTime": "2023-01-09 14:49:37",
    "systemVol": 100,
    "temperatureMode": 0,
    "threshold": 70,
    "throughType": "2",
    "warningTemperature": 37.200000762939453,
    "wifiMac": "94:A4:08:2E:79:17"
  },
  "dev_sno": "02:02:F4:33:47:05",
  "method": "get_basic_settings",
  "msg": "Get device basic Settings parameters ",
  "success": true,
  "token": "eyJkZXZpYTE1NjM3MzR9"
```

- Cloud returns:



Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

7.6. Update basic device Settings in the cloud

Purpose: Update basic device Settings

The device will reboot after setting

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value <code>set_basic_settings</code>
data	is	object	Data
data.basicSettings	is	object	Set parameters
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- basicSettings Description:

The cloud can obtain the basic Settings of the device. To set the parameters, you only need to deliver the parameters that need to be set.

- Example

```
{
  "method": "set_basic_settings",
  "data": {
    "dev_sno": " 04:12:53:FA:21:DA",
    "basicSettings": {
      "name": "Attendance machine ",
      "location": "Gate ",
      "rebootEveryday": false,
      "rebootTime": ""
    }
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
```

```

    "method": "set_basic_settings",
    "success": true,
    "msg": "Setup success ",
    "params": {}
}

```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```

{
  "code": 0,
  "msg": "OK",
  "success": true
}

```

7.7. Update personalization in the cloud

This interface parameter has been merged into `set_basic_settings`, but the interface is still available

The device will reboot after the Settings are set

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction, with the value <code>set_personal_settings</code>
data	is	object	Data
notify	is	string	The device returns the request address of the execution result



params	is	object	Reserved fields, notify required params parameters, provided by the cloud service
--------	----	--------	---

- personalSettings Description:

Parameter name	Required	Type	Instructions
picture	is	string	logo picture base64 string without prefix, see example
bannerEnable	is	bool	Open AD
bannerString	is	string	Advertising slogan
nameDisplay	is	bool	Identify the result, 0 unencrypted, 1 encrypted

- Example

```
{
  "method": "set_personal_settings",
  "data": {
    "dev_sno": " 04:12:53:FA:21:DA",
    "personalSettings": {
      "picture":
        "iVBORw0KGgoAAAANSUhEUgAAAEQAAABECAY ..." ,
      "bannerEnable": true,
      "bannerString": "XX",
      "nameDisplay": true
    }
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
----------------	----------	------	--------------



dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "set_personal_settings",
  "success": true,
  "msg": "Setup success ",
  "params": {}
}
```

- Cloud returns:

Parameter name	Required	type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
  "code": 0,
```

```

    "msg": "OK",
    "success": true
  }

```

7.8. The cloud sends a remote upgrade command

Purpose: device software upgrade

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction with the value device_upgrade
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

data parameter description:

Parameter name	Required	Type	Instructions
firmwareVersion	is	string	The current firmware version of the device, which is obtained by obtaining basic device parameters
model	is	string	Device model and platform are obtained from Obtaining Basic Device parameters
download_url	is	string	Firmware download address

- Example

```

{
  "method": "device_upgrade",
  "data": {
    "firmwareVersion": "xxx.xxx.xxx.xxx",
    "model": "Cxxx",
    "Download_url" : "http://192.168.1.16:5389/xxxxxx.img",
  },
}

```



```
"Notify" : "http://192.168.1.16:5389/device/notify",  
"params": {}  
}
```

Device requests feedback from the cloud on execution results:

- Request method: POST
- Content-Type: application/json
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

```
{  
  "dev_sno": "04:12:53:FA:21:DA",  
  "token": "38f8cc4bc6fb4e82c78175e2d56",  
  "method": "device_upgrade",  
  "success": ,  
  "msg": "upgrade instruction accepted successfully ",  
  "params": {}  
}
```

Cloud return:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform reception succeeds or fails

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

7.9. The cloud sends a remote restart command

Purpose: Remote restart device

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction with the value device_restart
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "method": "device_restart",
  "data": {},
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
----------------	----------	------	--------------



dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
{
  "dev_sno": "JWSZ1209323212",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "open_door",
  "success": true,
  "msg": "Restart successful ",
  "params": {}
}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```

8. Device time period control

8.1. The cloud sends a command to synchronize the common time range

Purpose: The platform is used to issue a common time period. Every time the platform issues a command, the device obtains the information of the corresponding time period of the platform (that is, the device only obtains the time period corresponding to the TimeGroupId).

Common time range is used to control the traffic of whitelisted personnel within a specified time range

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with value device_timeAccessGroups
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- data Parameter Description

Parameter names	Required	Type	Instructions
path	is	string	Request the cloud address of the access time period interface
path_params	is	object	Parameters to be carried by the access control period interface



- path_params Parameter Description

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device number

- Example

```
1. {  
2. "method":"device_timeAccessGroups",  
3. "data":{  
4. "path" : "http://192.168.1.16:5389/device/ device_timeAccessGroups",  
5. "path_params":{  
6. }
```

Device request **data.path** to obtain access control time group information:

Parameters

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device number
token	is	string	token
path_params	is	object	path_params provided by the platform issuer

data The contents of the message body

Data type	Field name	Instructions
List<WeekTime>	TimeGroup	

Definition of the WeekTime structure

Data types	Field name	Instructions
String	TimeGroupId	Time group Unique ID (Only 4 groups of time periods can be set) Fixed as 1, 2, 3, 4
List<DayTimes>	TimeGroup	

Definition of DayTimes structure

Data types	Field name	Instructions
String	WeekIndex	Corresponding weeks: Monday, 1 Tuesday, 2 Wednesday, 3 Thursday, 4



		Friday, 5 Saturday, 6 Sunday
List<TimesBean>	Times	

TimesBean structure definition

Data types	Field name	Instructions
String	Start	Start time format Example: 09:00
String	End	End format Example: 18:00

Return example

```
{
  "TimeGroup": [{
    "TimeGroup": [{
      "Times": [{
        "End": "02:00",
        "Start": "01:00"
      }],
      "WeekIndex": "0"
    },
    {
      "Times": [{
        "End": "02:00",
        "Start": "01:00"
      }],
      "WeekIndex": "1"
    }
  ]},
  "TimeGroupId": "2"
}]
}
```

8.2. The cloud sends a command to synchronize a special time period

Purpose: The platform is used to issue the special time range command. The special time range rule takes precedence over the common time range

The special time is used for whitelisted personnel to permit or deny traffic within this time range. The special time range rule takes precedence over the common time range.

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Name of the instruction, with the value device_AccessSpecialTimeGroup
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
SpecialTimeGroupId	is	int	Special Time Period id
params	is	object	Reserved field, notify required params parameter, provided by the cloud service

- data Parameter Description

Parameter names	Required	Type	Instructions
path	is	string	Request the cloud address of the access time period interface
path_params	is	object	Parameters to be carried by the access control period interface

- path_params Parameter Description

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device number

- Example

```

1. {
2. "method":"device_AccessSpecialTimeGroup",
3. "data":{
4. "path" : "http://192.168.1.16:5389/device/ device_AccessSpecialTimeGroup",
5. "path_params":{}
```

6. "timeGroupId ":1,
7.}

Device request **data.path** to obtain access control time group information:
argument

Parameter name	Required	Type	Instructions
path	is	string	Request URL
dev_sno	is	string	Device number
token	is	string	token
path_params	is	object	path_params provided by the platform issuer

The platform returns the contents of the body of the data message

Data type	Field name	Instructions
List<SpecialTimeGroupData>	SpecialTimeGroup	

Definition of the SpecialTimeGroupData structure

Data type	Field name	Instructions
String	SpecialTimeGroupId	Special Time Period ID (Less than or equal to 0, indicating no restricted access)
List<TimesBean>	SpecialTimeGroupData	

Definition of TimesBean structure

Data types	Field name	Instructions
long	Start	Represents the number of milliseconds of the start time
long	End	The number of milliseconds that represent the end time
int	PassStatus	0 Passstatus, 1 prohibit

Platform Return Example

```
{
  "SpecialTimeGroup": [{
    "SpecialTimeGroupData": [{
      "End": "1654931398000",
      "PassStatus": 0,
      "Start": "1654927798000"
    }]
  }].
```

```
        "SpecialTimeGroupId": "1"  
    } ]  
}
```

9. Compare records to upload

9.1. Upload the comparison records

Brief Description

- Device upload identification record
- The interface address is configured in the device configuration
- The snap score is the similarity of the person most similar to it

Request URI

- </record/upload/online>

Request method

- POST

Parameter types

- application/json

Parameters

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device number
token	is	string	token
cmp_type	is	int	Type of alignment, 0: other; 1: comparison between witness and witness; 2: face comparison
capture_img	is	string	Field capture photo, base64 encoding required



capture_time	is	string	Compare time, timestamp accurate to millisecond. See example below.
capture_score	is	string	Compare score values
capture_status	is	int	Int result, 0 match failed, 1 match passed
person_id	no	int	Person id
person_name	no	string	Name
person_temperature	no	float	Body temperature (Celsius)
person_type	no	string	Personnel type
id_card	no	string	Id card
ic_card	no	string	IC card number
qr	no	string	Qr Code
card_data	no	object	Data in the list or read ID data, ID information, etc
card_data.sex	no	int	Gender 0 Unknown 1 male 2 female
card_data.faceImg	no	string	Id photo, base64 encoded
card_data.identityId	no	string	Id number
card_data.signDepartment	no	string	Visa Authority
card_data.nation	no	string	Nationalities
card_data.address	no	string	Address



healthCode	no	int	0 not turned on, 1 green code, 2 yellow code, 3 red code
rna	no	object	Nucleic acid Test information
rna.ret	no	int	Nucleic acid test results (0 unknown, 1 negative, 2 positive)
rna.time	no	string	Nucleic acid test time
rna.src	no	string	Nucleic acid Testing Facility
antiboby.ret	no	string	Vaccinate or not 0 Not vaccinated, 1 vaccinated
antiboby. time	no	string	Inoculation time yyyy-mm-dd hh:mm:ss
antiboby. src	no	string	Vaccination records (Not yet vaccinated, 1 dose vaccinated, 2 doses vaccinated, fully vaccinated)

Sample request

```
{
  "dev_sno": " 04:12:53:FA:21:DA",
  "token": "f9k4l8ds3e92w02d9c82013na0s9",
  "person_id": 12306,
  "person_name": "Zhang SAN ",
  "person_temperature": 36.7,
  "person_type": "1",
  "id_card": "",
  "ic_card": "",
  "capture_img ": "/9i/4aVBORw ... .." ,
  "capture _score": "85.64",
  "capture_time": "1610967016235",
  "capture_status":1,
  "card_data" :
    {
      "address" : "",
      "antiboby" :
        {
          "ret" : 0,
```

```

        "src" : "",
        "time" : ""

    },
    "healthCode" : 0,
    "identityId" : "",
    "rna" :
    {
        "ret" : 0,
        "src" : "",
        "time" : ""

    },
    "sex" : 0,
    "signDepartment" : "",
    "validEndDate" : "",
    "validStartDate" : ""

    },
}

```

Platform return parameter description

Parameter name	Type	Instructions
code	int	Status code 0 Success non-0 failure
msg	string	Status Instructions
success	bool	Make or break

Return Example

```

{
  "code": 0,
  "msg": "OK",
  "success": true
}

```

9.2. Set device repush record

Purpose: The device pushes the recognition record again within the **set period** of time

Command parameters:



Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value setDeviceRecord_revert
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service
data.startTime	is	long	The start time of the record
data.endTime	is	long	The end time of the record

- Example

```
{
  "method": "setDeviceRecord_revert",
  "data": {
    "startTime": "",
    "endTime": ""
  },
  "Notify" : "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:

- Request method: **POST**
- Content-Type: **application/json**
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value



success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

- Example

```
1. {  
2. "dev_sno":"JWSZ1209323212",  
3. "token":"38f8cc4bc6fb4e82c78175e2d56",  
4. "method":"setDeviceRecord_revert",  
5. "success":true,  
6. "msg":" Setup success ",  
7. "params":{}  
8.}
```

- Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

- Example

```
{  
    "code": 0,  
    "msg": "OK",  
    "success": true  
}
```

9. Other comparison modes

9.1. Qr code recognition Settings

Purpose: Set two-dimensional code recognition (the content of two-dimensional code cannot be pure numbers, the length is greater than or equal to 16)

Command parameters:

Parameter name	Required	Type	Instructions
method	is	string	Instruction name with the value qr_mode
data	is	object	Data
notify	is	string	The device returns the request address of the execution result
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

data parameter description:

Parameter name	Required	Type	Instructions
qr_online	is	bool	true: online QR code comparison false: compare the offline QR code
qr_api	is	string	qrcode Identification address (http address)

- Example

```
{
  "method": "qr_mode",
  "data": {
    "qr_online": true,
    "qr_api": "http://xxxxxxxxx"
  },
  "Notify": "http://192.168.1.16:5389/device/notify",
  "params": {}
}
```

Device requests feedback from the cloud on execution results:



- Request method: POST
- Content-Type: application/json
- Request parameters:

Parameter name	Required	Type	Instructions
dev_sno	is	string	Device MAC
token	is	string	token value
success	is	bool	Make or break
method	is	string	Instruction name
msg	is	string	Instructions
params	is	object	Reserved fields, notify required params parameters, provided by the cloud service

Sample request:

```
{
  "dev_sno": "04:12:53:FA:21:DA",
  "token": "38f8cc4bc6fb4e82c78175e2d56",
  "method": "qr_mode",
  "success": true,
  "msg": "Setup success ",
  "params": {}
}
```

Cloud returns:

Parameter name	Required	Type	Instructions
code	is	int	Status code, 0 succeeded, non 0 exception
msg	is	string	Instructions
success	is	bool	Platform receive success or failure

Return an example:

```
{
  "code": 0,
  "msg": "OK",
  "success": true
}
```



9.2. Qr code online recognition

Purpose: The device recognizes the two-dimensional code information in real time, requests the platform, and whether the platform returns to open the door,

Used for visitor two-dimensional code recognition

The two-dimensional code recognition interface can be run separately from the entire cloud protocol, and the configuration tool can configure the online comparison address

Device-side request

- Request parameters:

The device requests the qr_api address and pushes the two-dimensional code to the platform to judge the validity

Parameter Name	Required	Type	Instructions
qrcode	is	string	Scanned two-dimensional code string
deviceSN	is	string	Device SN
dev_sno	is	string	Device number (device MAC address)
time	is	string	Scan time yyyy-MM-dd hh:mm:ss

Sample request:

```
{
  "qrcode": "",
  "time": "2021-06-09 12:00:00"
}
```

Platform return

Parameter name	Required	Type	Instructions
name	no	string	Name
code	is	int	0 success not 0 failure
message	no	string	Identifying information

Return example:

```
{
  "name": "",
  "code": 0,
  "message": "QR code comparison is successful"
}
```

```
}
```

9.3. Qr code offline recognition

The offline recognition of the two-dimensional code is to compare the qr in the sender, and the QR content can be delivered to the qr field when the person synchronizes.

9.4. Online comparison and identification

Purpose: Push the data to the platform address after device identification, and return the information to the device for door opening and prompt after recognition and judgment

The online interface can be run independently from the whole cloud protocol and LAN, the token is not the key information, and the configuration tool can directly switch the comparison mode and configure the online comparison address.

This interface can be understood as a separate set of protocols

Request URI

- [/api/OnlineComparison/](#)

(When used in conjunction with the cloud protocol, the requested address for online alignment is url+/api/OnlineComparison, and when used alone, the requested address can be configured directly through the tool)

Method of request

http post

Device Request parameters:

Parameters	Parameter types	Must not	or	Paraphrase
deviceId	string	is		Device Unique Identification (mac)
dev_sno	string	is		Device number (Device MAC address)
deviceSN	string	is		Device SN
token	string	no		token
type	string	is		How to identify: type= stranger, which stands for stranger type=face, face recognition; type=qr, representing scan code recognition; type=id, which represents the comparison between witnesses and witnesses;



			type=ic, representing swipe card comparison;
value	string	is	Identify the content: When type= stranger, the value content is the fixed character "data.capture" When type=face, the value content is the person id number; When type=qr, the value is the content string of the recognized two-dimensional code. When type=id, the value content is the ID card number recognized by the ID card;
temperature	string	no	Recognize temperature values and use them as background records only;
data	object	no	Data
data.capture	string		Live snap photo, base64 encoded
data.time	string		Identify the time, with a timestamp accurate to milliseconds

Sample request:

```
{
  "dev_sno": "04:0C:F4:12:68:05",
  "deviceId": "0A:0C:F4:12:68:05",
  "devicesN": "T02F11601FS",
  "temperature": "",
  "token": "dc70ace460a00a093b51efff3588d47b",
  "type": "",
  "value": "",
  "data": {
    "capture": "",
    "datetime": ""
  }
}
```

Platform return argument:

Parameters	Parameter type	Paraphrase
status	int	0: successful; 1: failure (non-zero failure). Open the door on success, not on failure.



msg	String	Prompt message. If the content is returned, it is displayed, if it is not returned, it is not displayed.
audio	String	Voice (need to use smart voice module, voice module connected to serial port 2, this field is not applicable to T12)

```
{
  "status": 0,
  "msg": "Recognition success",
  "audio": "Identification success"
}
```

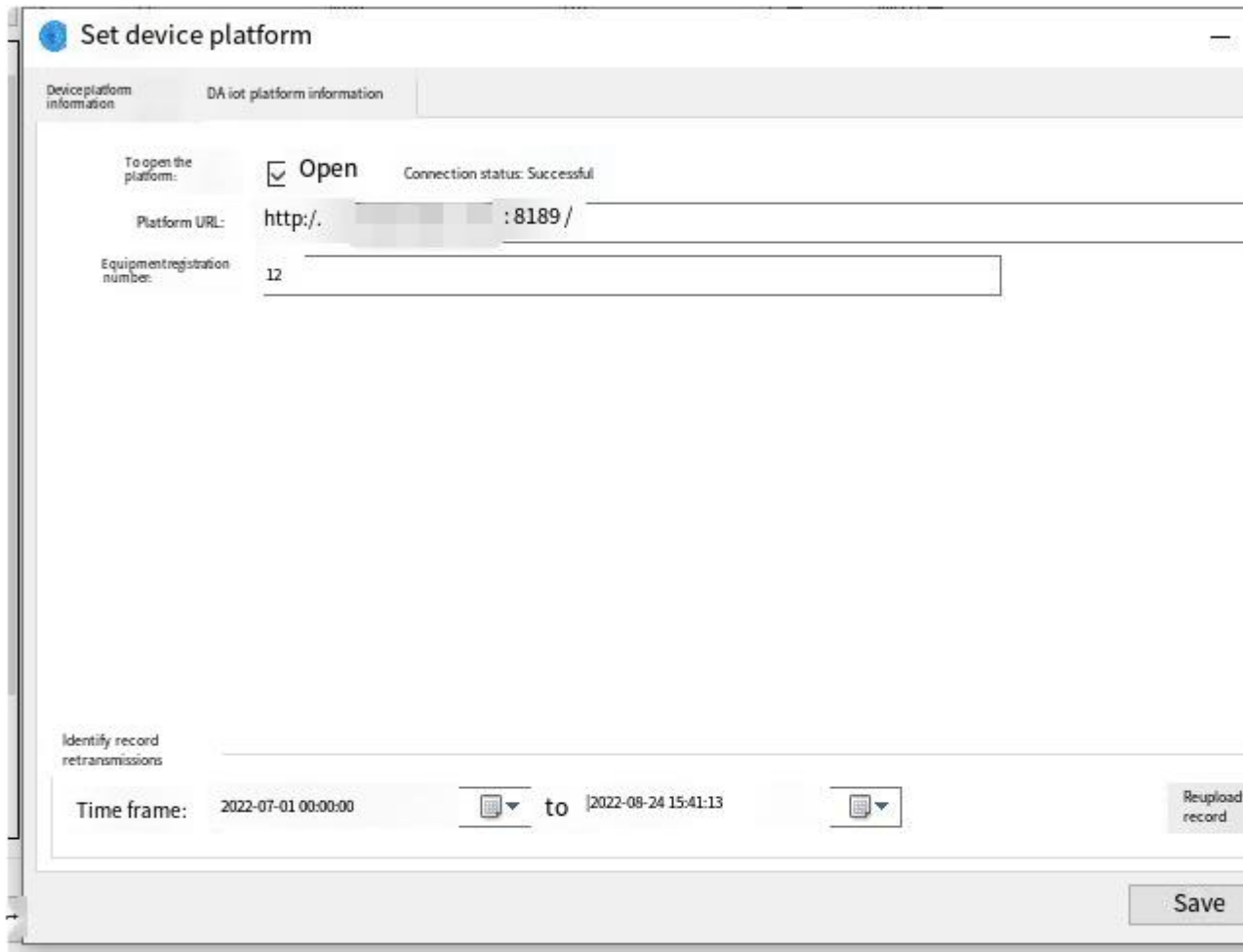
Attachment 1. Device configuration connection description

Tools download address <http://47.104.64.115:9001/>



Set the device platform as shown below:

This platform address is the platform address of the peer party. (Device registration number as long as it is not empty)



Enter the platform address: http://[redacted]:8189/

Annex 2. The platform does not want to do too much functional development, just want to identify the record (traffic data) processing method

Key point: The project does not use the [face recognition intelligent management system] we provide, that is, the project can not call the "LAN http docking protocol" client software, otherwise this method is invalid

<https://shimo.im/docs/WIArzGboORSzvZA2/> "LAN docking HTTP protocol"

The interface address and url pushed by this protocol can be configured according to the positions in the screenshot (the ip address and url in the screenshot are examples and have nothing to do with the actual development)

Parameter setting

Basic parameters Data upload UI Settings Logo Settings Standby Settings Health code Settings

Device IP address: 192.168.2.200

Device MAC address: 0A:0C:FO:12:2F:D1

Note: Heartbeat and history push data according to the LAN protocol. Please fill the heartbeat setting according to the protocol when filling in the API address

Api address:

History push Settings

Api address:

http://192.168.1.200:8081/getrecord/

Identify record
retransmissions

Time frame:

2022-11-22 14:14:40



to

2022-11-24 14:14:40



Video Preview

Face release

Aging test

Clear device list

Set the device platform

Modify device parameters

Bulk sa

Get a singl

Request method: post

The contents of the message body:

Data type	Field name	Required	Instructions
String	id	Y	uuid
String	Mac_addr	Y	Device unique identification code
String	time	Y	Compare time yyyy-MM-dd HH:mm:ss
String	devicename	N	Device name
String	location	N	Installation location



int	inout	N	Access 0 Exit 1 entrance
String	employee_number	Y	Person number (blank for strangers)
String	name	N	Name
String	sex	N	Gender
String	nation	N	Nationality
String	idNum	N	Id number
String	icNum	N	IC card number
String	birthday	N	Date of birth
String	telephone	N	Phone
String	address	N	Address information on your ID card
String	depart	N	Issuing authority
String	validStart	N	Start of expiration date
String	validEnd	N	Expiration date
int	IdentifyType	Y	Identification method (comparison type) : 0 face recognition, 1 blacklist recognition (reserved field), 2 witness comparison, 3 IC card recognition
int	resultStatus	Y	Comparison result 1 Successful comparison 0 failed comparison
String	face_base64	Y	Compare the snap photo base64 bit string
String	templatePhoto	N	Stencil photo
String	temperature	N	Body temperature
int	healthCode	N	Health Code Type 0 Disabled 1 Green code 2 Yellow code 3 Red code
json	rna	N	Other test information
Int	rna.ret	N	Nucleic acid test results (0 not turned on, 1 negative, 2 positive)
String	rna.time	N	Nucleic acid test time
String	antiboby.ret	N	Vaccinate or not 0 Not vaccinated, 1 vaccinated
String	antiboby. time	N	Inoculation time yyyy-mm-dd hh:mm:ss
String	antiboby. src	N	Vaccination records

The platform must reply after the device pushes the record

Normal return: {"message": "Set access groups success", "result": 0}

Abnormal return: {"message": "Set access groups fail", "result": error code}



Annex III. Data encryption transmission

See LAN Protocol 2.9 Device Data Transmission Encryption below

<https://shimo.im/docs/WIArzGboORSzvZA2>

LAN protocol

If cloud protocol does not apply, check LAN protocol

<https://shimo.im/docs/WIArzGboORSzvZA2>